

Evangelis Silva

Department of Computer Science, University at Albany, SUNY
Albany, NY 12222

esilva2@albany.edu | +1 (518) 961-5959
evangelisilva.github.io | GitHub: evangelisilva | LinkedIn: evangelisilva

Education

University at Albany, SUNY

January 2022–Present

Ph.D. in Computer Science, in progress

Advisor: Prof. Jeong-Hyon Hwang

Research focus: Database and storage systems

University of Moratuwa, Sri Lanka

January 2014–March 2019

B.Sc. (Hons.) in Information Technology & Management

GPA: 3.50/4.20 | Advisor: Dr. Supunmali Ahangama

Research Interests

Database and storage systems; on-disk index structures; memory-efficient indexing and caching; experimental evaluation of data-intensive systems.

Publications

E. Silva, M. Nandathilaka, S. Dalugoda, T. Amarasinghe, S. Ahangama, and G. T. Weerasuriya. “Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images.” *Proceedings of the 6th International Conference on Information Technology Research (ICITR)*, IEEE, 2021.

Best Paper Award, AI & ML Track.

Research Experience

Doctoral Research — University at Albany, SUNY

January 2022–Present

Advisor: Prof. Jeong-Hyon Hwang

On-Disk Index Design and Evaluation

- Investigate an alternative on-disk indexing approach intended to reduce page-management overhead and improve memory utilization for key-value storage.
- Developed a benchmarking framework with common index interfaces, workload generators, and performance measurements to support fair and reproducible comparisons.
- Conducted preliminary evaluations of B+-trees, LSM-trees, and Bf-Tree across dataset scales, key/value configurations, memory budgets, insertion orders, access patterns, workload types, and storage media.
- Analyze operation latency and storage I/O behavior to characterize performance trade-offs; obtained preliminary results indicating potential benefits of the proposed approach.
- Preparing a benchmark manuscript describing the framework and preliminary findings.

An Experimental Evaluation of Group Farness Estimation Techniques

- Conducted an empirical study of scalable group farness estimation in large graphs, extending single-source algorithms to multi-node group settings.
- Evaluated sampling and cardinality estimation methods, including KMV and Cohen’s HIP, to analyze accuracy, memory consumption, and runtime trade-offs.

Teaching Experience

Graduate Teaching Assistant

January 2022–Present

University at Albany, SUNY

- Conduct laboratory sessions and lead discussions to reinforce course concepts and guide students through programming and problem-solving exercises.
- Develop quizzes and grading rubrics to assess student understanding and support consistent evaluation.
- Serve as Head Teaching Assistant for selected courses, coordinating teaching assistants and graders and overseeing grading activities.
- Deliver lectures and conduct classes when covering for course instructors.
- Hold office hours and provide individual support with course material, assignments, and debugging.
- Assess student work and provide constructive feedback to strengthen conceptual understanding and technical skills.

Courses Supported

Course	Title	Semesters
CSI 213	Data Structures	Fall 2022; Fall 2023; Spring 2024; Spring 2025; Spring 2026; Fall 2026
CSI 410	Database Systems	Spring 2022; Fall 2022; Fall 2024; Fall 2025; Spring 2026
CSI 404	Computer Architecture and Organization	Spring 2022; Fall 2026
CSI 508	Database Systems I (graduate)	Spring 2022
CSI 210	Discrete Structures	Fall 2024

Honors and Awards

- **Karen R. Hitchcock New Frontiers Fund Award**, Initiatives for Women, University at Albany, 2026. \$1,000 to support research and professional development.
- **Best Paper Award, AI & ML Track**, ICITR, 2021, for “Machine Learning-Based Automated Tool to Detect Sinhala Hate Speech in Images.”
- **Customer First Value Award**, Sysco LABS, 2020.
- **Dean’s List Recognition**, University of Moratuwa, Level 2, Semester 2.

Academic Service and Outreach

- **Conference Paper Reviewer**, 36th International Conference on Scientific and Statistical Database Management (SSDBM), 2024.
- **Database Session Tutor**, Undergraduate Women Coding Club Bootcamp for IEEE Student Chapter Hackathon Preparation, University at Albany.
- **Technical Coordinator and Article Reviewer**, fit2school, Faculty of Information Technology outreach project, University of Moratuwa.

Additional Leadership

- **Student Relations Lead**, Vesess Alpha Coding Challenge.
- **Treasurer**, Sysco LABS SL Toastmasters Club.
- **Editor**, *FIT Chronicles*, Faculty of Information Technology newsletter, University of Moratuwa, 2017.
- **Organizing Committee Vice President (Delegate)**, IT Recruitment 2016, AIESEC Colombo South.

Industry Experience

Senior Software Engineer — Sysco LABS, Sri Lanka

November 2020–December 2021

- Designed and scaled distributed microservice systems supporting Sysco’s digital platforms.
- Improved query performance through indexing and caching optimizations, reducing average latency from 400 ms to 100 ms.
- Implemented real-time data synchronization between legacy and microservice databases using Change Data Capture (CDC) and AWS Kinesis.
- Developed RESTful APIs and data pipelines across regional clusters; supported reliability through monitoring and distributed tracing.

Software Engineer — Sysco LABS, Sri Lanka

March 2019–November 2020

- Developed backend services and RESTful APIs for Sysco’s global digital platform.
- Implemented CI/CD pipelines with Jenkins and Concourse and enhanced service observability using Datadog, Kibana, and Amazon CloudWatch.
- Optimized microservice data flows and collaborated with teams in the United States and Canada to maintain and scale production systems.

Engineering Intern (Data Analytics) — Vesess, Sri Lanka

October 2017–March 2018

- Developed automated data collection and enrichment pipelines for the Donordo analytics platform using Python and web scraping frameworks.
- Built data processing workflows and visual analytics dashboards to support donor-matching analysis.

Selected Systems and Software Projects

Mimix Data Transformer

- Developed a modular Java library for event stream transformation, mapping, and post-processing across relational and streaming data systems.
- Designed a pluggable architecture with filters, transformation hooks, and custom database writers; integrated AWS Kinesis and packaged the library as a Maven-compatible artifact.

Enterprise Email Service

- Designed and developed an internal email platform used by more than 20 enterprise services, supporting automated, template-based communications.
- Built RESTful APIs and optimized database schemas for reliable message delivery and integration with legacy systems.

Compassionate Sri Lanka

- Contributed to a nationwide donor–recipient platform in collaboration with the Ministry of Social Empowerment.
- Developed backend services and a Node.js/AngularJS application with SMS and email notifications for donation tracking.

Technical and Research Skills

Languages: Java, Python, C++, Rust, JavaScript, SQL, Bash.

Databases: PostgreSQL, RocksDB, MongoDB, Amazon DynamoDB, Redis.

Systems and tools: AWS, Docker, Spring Boot, Node.js, Git, Maven, Gradle, Jenkins, Concourse CI.

Research methods: Experimental evaluation, benchmarking, storage I/O analysis, system design and implementation, data visualization.

Analysis and writing: NumPy, Pandas, Matplotlib, scikit-learn, L^AT_EX.